Advanced Machine Learning, ETH Zurich

Convolutional filters

Carlos Cotrini

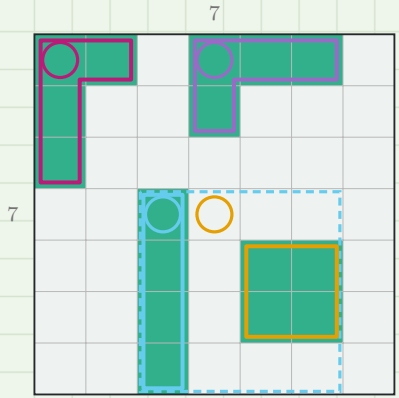
Autumn semester 2026

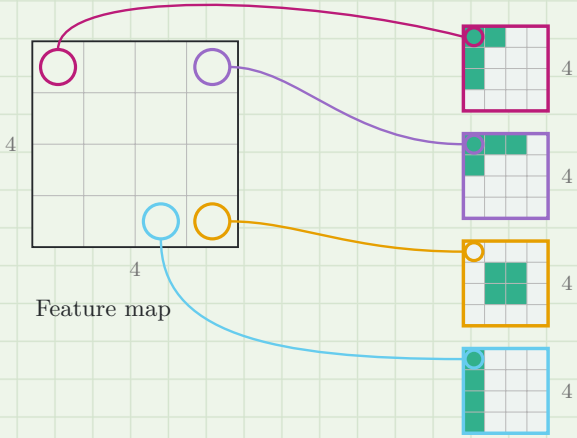
Subject

Decomposing an image into tiles

Entry

1





Subject

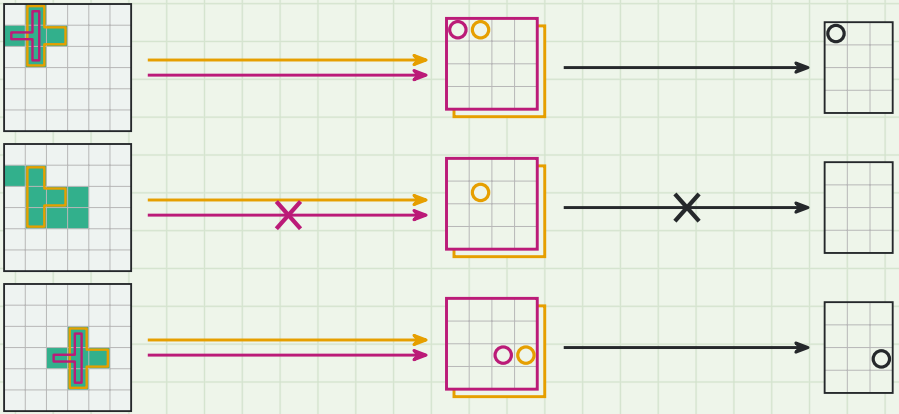
Recognizing islands that look like ✚

Entry

2









0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	1	1	0
0	0	0	1	0	0
0	0	0	0	0	0



1	0	0
1	1	0
1	0	0

3×3

0	0	0		1	0	0		0	0	0	sum	0
0	0	0	$*$	1	1	0	$=$	0	0	0	$\rightarrow$	
0	0	0		1	0	0		0	0	0		

0	0	0		1	0	0		0	0	0	sum	2
1	0	0	$*$	1	1	0	$=$	1	0	0	$\rightarrow$	
1	1	0		1	0	0		1	0	0		

1	0	0		1	0	0		1	0	0	sum	4
1	1	0	$*$	1	1	0	$=$	1	1	0	$\rightarrow$	
1	0	0		1	0	0		1	0	0		

Convolution

0	0	0	1
0	0	2	2
0	1	2	4
0	0	2	2

$\xrightarrow{-3}$

−3	−3	−3	−2
−3	−3	−1	−1
−3	−2	−1	1
−3	−3	−1	−1

$\xrightarrow{\text{ReLU}}$

Desired output

0	0	0	0
0	0	0	0
0	0	0	1
0	0	0	0

Subject

Recognizing islands that look like ✚, with pooling

Entry

4

